

Super-Resolution for Satellite Imagery using Enhanced SRGAN

A report submitted by

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I, the undersigned, hereby declare that this thesis is my own original work. All information and ideas from other sources have been properly cited and acknowledged. The work presented is free from plagiarism and fabrication. I understand that any violation of academic integrity may result in disciplinary action by the Institute.

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Approval Sheet

This report, titled “Super-Resolution for Satellite Imagery using Enhanced SRGAN,” is submitted by Bhargav Pipaliya (Enrollment No: 221040011003) in partial fulfillment of the requirements for the Bachelor of Technology degree.

Place:

Date:

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The committee reviewing this report confirms that the report titled “Super-Resolution for Satellite Imagery using Enhanced SRGAN,” authored by Bhargav Pipaliya, meets the academic and research standards required for submission to the Department of Computer Engineering at the Institute of Infrastructure Technology Research and Management, Ahmedabad.

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Signature with date:

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Dedication

This work is dedicated to the Almighty, whose blessings provided the strength to complete this research. I also dedicate this thesis to my supervisor, Professor Ashish Soni, for his exceptional guidance and unwavering support, which were key to the success of this project. Finally, this work is dedicated to the researchers and scientists in the field of satellite image enhancement. Their contributions have inspired and motivated this exploration.

Abstract

High-quality satellite imagery is essential for applications ranging from environmental monitoring to urban planning. However, much of the available data is of low resolution, which limits its usefulness. This thesis explores the use of a deep learning technique called Super-Resolution to enhance these images. Specifically, it focuses on implementing and evaluating the Enhanced Super-Resolution Generative Adversarial Network (ESRGAN) on the OLI-2 MSI satellite dataset. The project involved preparing the satellite data, training the ESRGAN model to increase image resolution by a factor of four, and then testing its performance. The quality of the enhanced images was measured using standard metrics like the Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM). The results show that ESRGAN can significantly improve the clarity and detail of satellite images. It achieved an average PSNR of 33.64 dB and an SSIM of 0.87. This indicates a high degree of accuracy and structural fidelity in the generated images. This study demonstrates that advanced deep learning models like ESRGAN are powerful tools for improving the quality of satellite data, making it more useful for scientific analysis and decision-making.

Keywords: Super-resolution, GANs, ESRGAN, satellite images, deep learning, PSNR, SSIM

1 Introduction

Satellite images are very important for many areas, like watching the environment, planning cities development, farming, handling disasters, and defense. Being able to see clear, high-quality images help in understanding things better and making good decisions. But because of camera limits, weather, and costs, many satellite images are taken at lower quality. This makes them less useful for tasks that need very precise details. This project aims to solve this problem by using and testing ESRGAN (Enhanced Super-Resolution Generative Adversarial Network). This is an advanced computer model that makes blurry images clearer. It uses a special kind of artificial intelligence called Generative Adversarial Networks (GANs) to create clear, high-quality images from blurry ones. The goal is to make images much sharper and more real-looking than older methods.

1.1 Project Details and Image Information

In this project, ESRGAN was used with the OLI-2 MSI dataset. This dataset contains many satellite images that show Earth's land surface. This makes it very useful for studying remote areas. The dataset has both very clear (High-Resolution or HR) images and less clear (Low-Resolution or LR) images. The clear images were used as the correct versions. Here are the typical details for the images that were processed:

Clear (High-Resolution) Image Details:

- Image Size: 480 pixels wide × 480 pixels high
- Color Channels (Bands): 3 (like Red, Green, Blue)
- Data Type for Colors: float32
- Location System (CRS): EPSG:32649
- Resolution: 10 meters

Blurry (Low-Resolution) Image Details:

- Image Size: 160 pixels wide × 160 pixels high
- Color Channels (Bands): 3
- Data Type for Colors: float32
- Location System (CRS): EPSG:32649
- Resolution: 30 meters

Looking at these details, the clear images are 480 × 480 pixels, and the blurry images are 160 × 160 pixels. This means that ESRGAN needs to make the blurry images 3 times clearer (480 divided by 160 equals 3). However, a direct 3x upsampling is not easily performable in the current scenario. Therefore, the images were further downsampled from 160 × 160 to 120 × 120, allowing for a proper 4x upsampling. This adjustment ensures that when ESRGAN performs 4x upsampling, the output matches the original 480 × 480 clear image size. This was important for training the model correctly and checking its quality. The main goal was to see how well ESRGAN could make satellite images clearer, keep small details, and improve their look. Standard image quality scores like PSNR and SSIM were used to measure this.

2 ESRGAN: How it Makes Images Clearer

2.1 What is ESRGAN?

ESRGAN stands for Enhanced Super-Resolution Generative Adversarial Network. It was created by Wang and others in 2018. It is a much better version of an older model called SRGAN. SRGAN was good, but sometimes it made images with weird spots or textures that did not look quite real. ESRGAN fixes these problems by changing how it is built and how it learns. This makes it one of the best models for making images look clearer and more natural. ESRGAN was chosen for this project because it is known to make high-quality images with finer details and more natural textures.

2.2 How ESRGAN is Built

ESRGAN has two main parts that work against each other: a Generator and a Discriminator. They are built to make images as clear and detailed as possible.

2.2.1 The Generator Network

The Generator is like an artist. Its job is to take a blurry (low-resolution) image and turn it into a clear (high-resolution) one. Its main features are:

- **Special Building Blocks (RRDBs):** The most important part of the Generator is called Residual-in-Residual Dense Blocks (RRDBs). These can be thought of as super-efficient mini-factories inside the artist. Unlike simpler blocks in older models, RRDBs are made of many smaller, connected parts (layers). This design helps the Generator go deeper, reuse ideas, and learn smoothly.
- **No Batch Normalization (BN):** In ESRGAN, Batch Normalization was removed from the Generator. This is because BN could sometimes cause strange patterns (artifacts) in the final images. Taking it out helps remove unwanted spots and makes the program more flexible on new images.
- **Upscaling Layers:** After the RRDBs do their work, special layers called Pixel Shuffle layers make the image bigger. These layers are very good at increasing the image size without adding blurry or blocky parts.

2.2.2 The Discriminator Network

The Discriminator is like a judge. Its job is to look at an image and decide if it is a “real” high-quality image or a “fake” one made by the Generator. Its main goal is to give strong feedback to the Generator, pushing it to make better images. Its main feature is:

- **Relativistic Average GAN (RaGAN) Judge:** ESRGAN uses a special kind of judge called RaGAN. A normal judge just says “real” or “fake.” But the RaGAN judge tries to decide if a real image is more real than a fake one. This makes the Generator try even harder to create images that are super convincing.

2.3 How ESRGAN Learns (Loss Functions)

To make images sharper and more real, ESRGAN uses a smart mix of “loss functions.” These are like scores that tell the model how well it is doing and what it needs to improve.

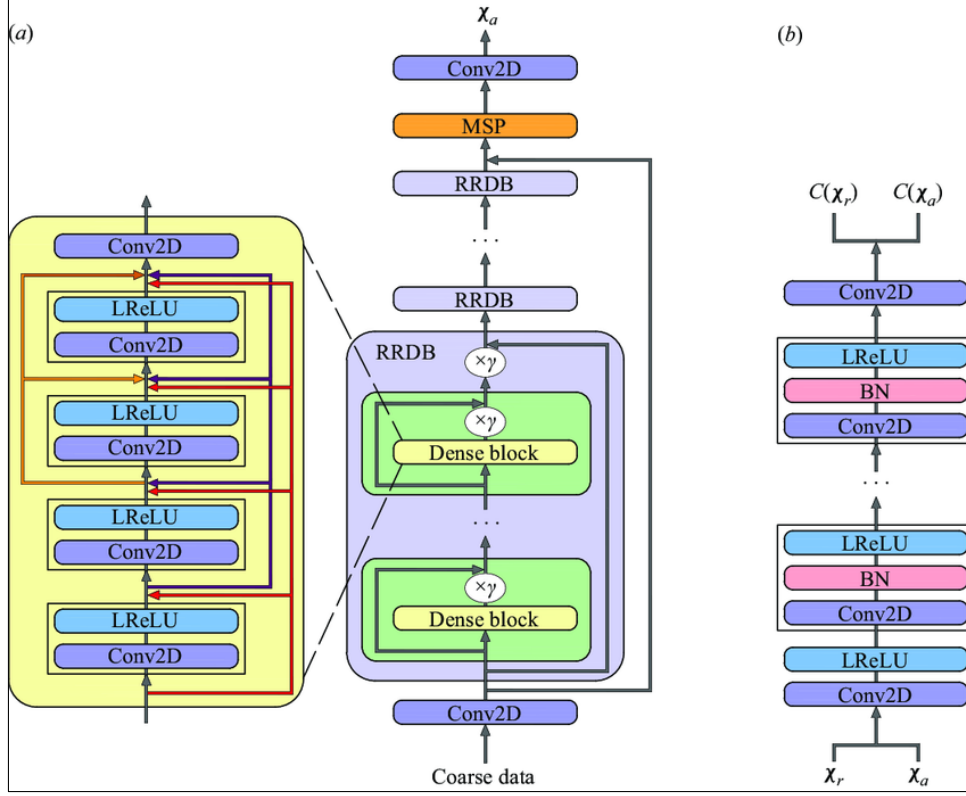


Figure 1: Architecture of ESRGAN. (a) The generator, where γ is the residual scaling parameter (set to 0.2), and (b) the discriminator.

2.3.1 Perceptual Loss

This score makes sure the generated image has similar high-level patterns and details as the original clear image. It uses a pre-trained VGG network to check features before the activation layers, which gives a richer comparison. This is very important for making textures look real.

2.3.2 Adversarial Loss

This score pushes the Generator to create images that are so real-looking that the Discriminator cannot tell them apart from actual high-resolution images. This helps bring back small details and textures. The Relativistic Average GAN (RaGAN) loss is defined as:

$$L_D^{Ra} = -E_{x_r} [\log (D_{Ra}(x_r, x_f))] - E_{x_f} [\log (1 - D_{Ra}(x_f, x_r))]$$

Where x_r is a real image, x_f is a fake (generated) image, and D_{Ra} is the relativistic discriminator's output.

2.3.3 Content Loss

This score makes sure that the generated image is very close to the original clear image, pixel by pixel. ESRGAN often uses L1 loss, which tends to make images less blurry compared to

the L2 loss used in older models. It provides the basic structure for the clearer image.

$$L_{\text{content}} = \|G(I_{LR}) - I_{HR}\|_1 \quad (2)$$

Where $G(I_{LR})$ is the generated super-resolved image, I_{LR} is the low-resolution input, I_{HR} is the ground-truth high-resolution image, and $\|\cdot\|_1$ denotes the L1 norm.

All these scores are combined to guide the Generator. The total loss for the Generator is a weighted sum of the perceptual loss, adversarial loss, and content loss. This smart combination helps ESRGAN avoid making images too smooth, makes them sharper, and improves their textures.

3 Project Work and Findings

3.1 Building ESRGAN for the Project

For this project, the standard ESRGAN design was used. The main parts of the network (like RRDBs and removing BN) were not changed. The main goal was to see how well the existing ESRGAN model works on satellite images. This design was kept because it is already proven to be good at making images look better.

3.1.1 Data Preparation and Image Scaling

A very important first step was preparing the OLI-2 MSI dataset. The clear (HR) satellite images were 480×480 pixels, and the blurry (LR) images were initially 160×160 pixels. To allow for a standard 4x upscaling, the LR images were downsampled from 160×160 to 120×120 pixels. This ensured the upsampled output would match the 480×480 size of the original clear images. The steps for data preparation involved reading images using the ‘rasterio’ tool to preserve important data and making sizes consistent for batch processing. This careful data preparation was key to a successful training process.

3.1.2 Tools and Libraries Used

A strong set of Python libraries was used to build and train the ESRGAN model:

- Python: The main programming language for the project.
- PyTorch: Used for building, training, and testing the deep learning model.
- Rasterio: Essential for reading and writing the special TIFF satellite images.
- OpenCV (cv2): Used for general image tasks, like resizing and downsampling.
- NumPy: Used for handling image data as numerical arrays.
- scikit-image: Provided tools for calculating image quality scores like PSNR and SSIM.
- Matplotlib/Seaborn: Used to create plots and visualize images.

3.2 Challenges During Training and Observations

During the training of the ESRGAN model, some common problems were encountered.

3.2.1 Training Instability

GANs can be hard to train because the Generator and Discriminator are always competing. It is tough to find a good balance where both improve together. Careful adjustments to the learning rate and other parameters were needed to ensure the model learned steadily. Unstable training sometimes resulted in the generator producing unusable “trash” images, as shown in Figure 2.

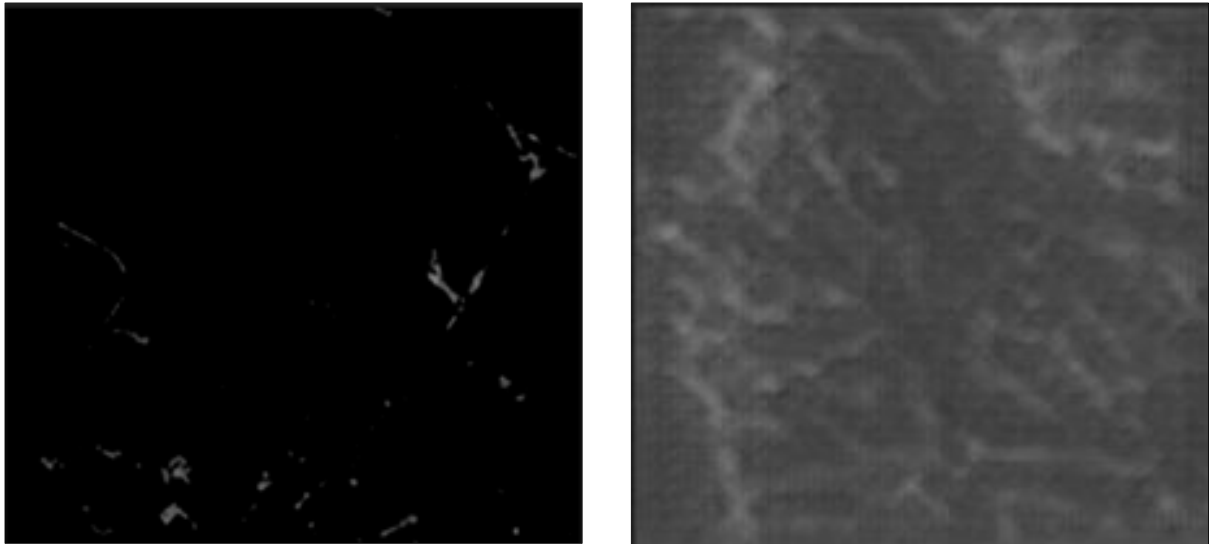


Figure 2: The generated trash images due to training instability.

3.2.2 Observation: The “Golden Yellow” Problem

During training and final image review, it was noticed that some of the generated images had a “golden yellow” tint (Figure 3). This color issue was traced back to a small error in the image preprocessing script. There was a problem with how the color values were handled, which caused a slight, unwanted color shift. The ESRGAN model then learned and amplified this color shift. The problem was fixed by correcting the image preprocessing script to ensure all color channels were handled correctly without adding any color bias. This experience showed how important perfect data preparation is in deep learning projects.

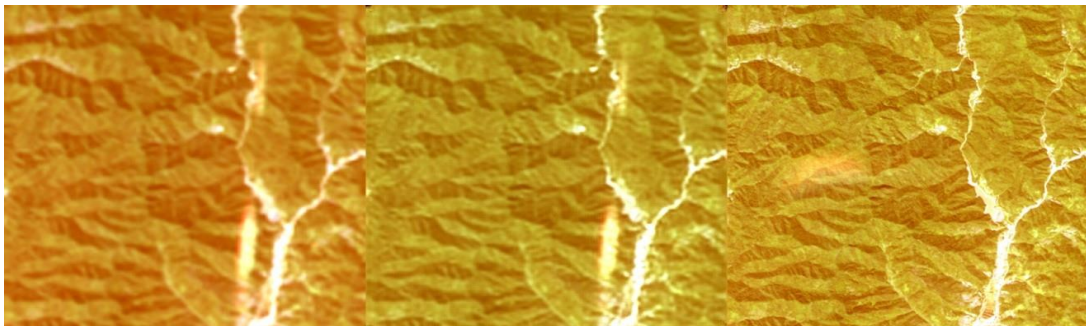


Figure 3: Example of a generated image showing the “golden yellow” color tint problem

4 Results

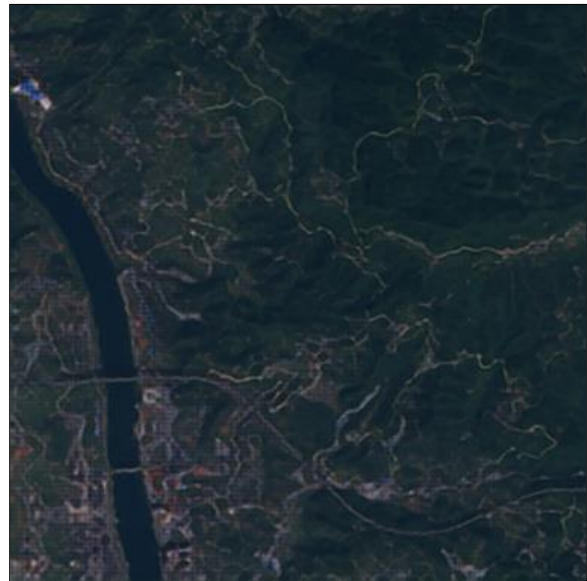
After extensive training and testing with the OLI-2 MSI dataset, the ESRGAN model showed excellent results in enhancing satellite images. The following metrics were obtained from testing over 100 images:

- Average PSNR over 100 images: **33.6426 dB**
- Average SSIM over 100 images: **0.8721**

These PSNR and SSIM values are very good. A PSNR above 33 dB means the generated images are very close to the original clear images. An SSIM of 0.8721 (close to 1) means they look very similar to the human eye. This shows that the model successfully preserved the original details and structure of the images.



(a)



(b)



(a)



(b)

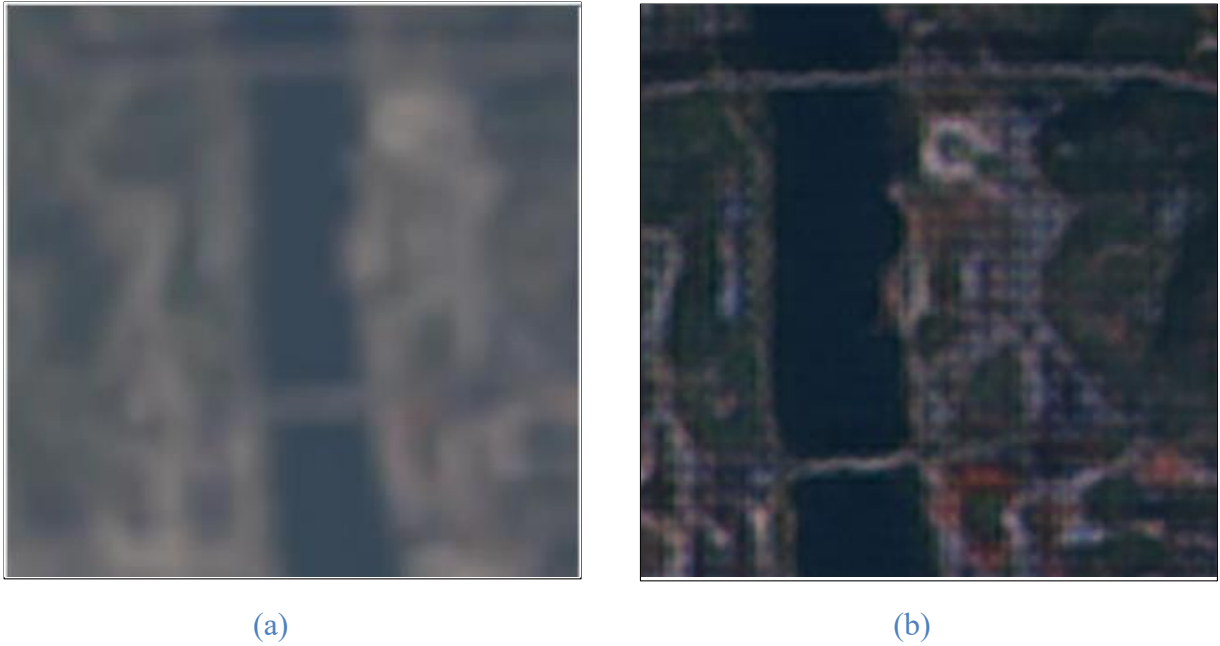


Figure 4: Detailed comparisons between (a): the original LR images (left in each pair) and (b): the generated SR images (right in each pair).

5 Future Work

While this project successfully demonstrated the effectiveness of ESRGAN in enhancing satellite imagery, several opportunities remain for further improvement and application. Future work could involve optimizing the model by fine-tuning hyperparameters and exploring alternative training strategies to further enhance PSNR and SSIM scores. Incorporating advanced degradation models that simulate real-world effects such as atmospheric noise and sensor blur could improve the model’s generalizability to raw satellite data. Additionally, designing a customized ESRGAN architecture tailored to the unique spectral characteristics of satellite images may yield better performance. Expanding the training dataset to include other satellite sources such as Sentinel-2 or Planet Scope would help create a more robust and versatile model. For practical deployment, techniques such as model quantization and pruning could be explored to enable real-time inference on edge devices or cloud platforms. Finally, developing a user-friendly graphical interface would make the system more accessible to researchers and analysts, facilitating broader adoption of the technology.

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Appendix:

Code Repository: <https://github.com/Bhargu9/Oli2MSI-ESRGAN>